

58. Temperature Sensor (TEMPS)

58.1 Overview

This MCU has a built-in temperature sensor. The temperature sensor outputs a voltage proportional to temperature. By converting the output voltage of the temperature sensor into a digital value with a 12-bit A/D converter unit 1 and converting it into temperature, the internal temperature of the MCU can be obtained.

Table 58.1 lists the specifications of the temperature sensor, and Figure 58.1 shows a block diagram of the temperature sensor.

Table 58.1 Specifications of Temperature Sensor

Item	Description
Temperature sensor voltage output	Temperature sensor outputs a voltage to the 12-bit A/D converter unit 1.
Low-power consumption function	Module stop state can be set.
Temperature Sensor Calibration Data	Reference data measured for each chip at factory shipment is stored.

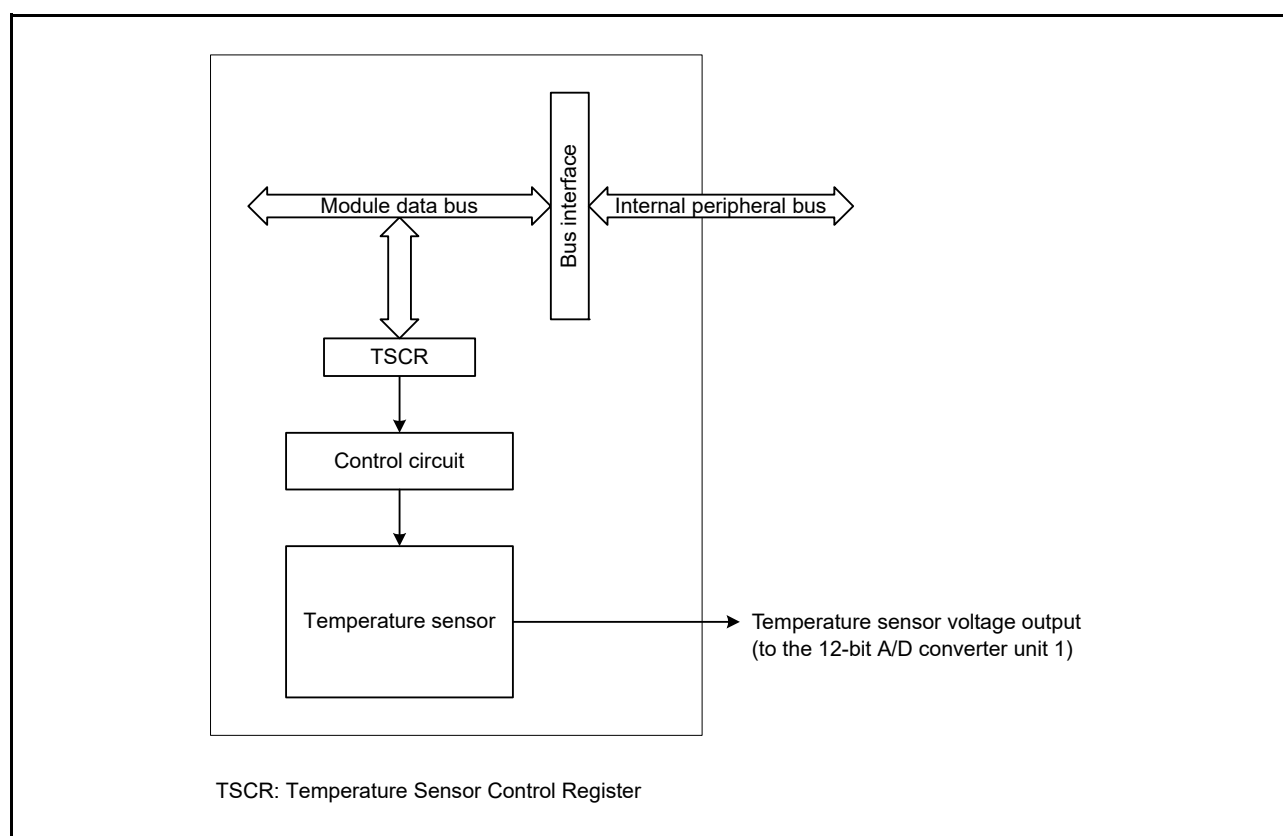


Figure 58.1 Block Diagram of Temperature Sensor

58.2 Register Descriptions

58.2.1 Temperature Sensor Control Register (TSCR)

Address(es): TEMPS.TSCR 0008 C500h

	b7	b6	b5	b4	b3	b2	b1	b0
	TSEN	—	—	TSOE	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b4	TSOE	Temperature Sensor Output Enable	0: Disables output from the temperature sensor to the 12-bit A/D converter unit 1. 1: Enables output from the temperature sensor to the 12-bit A/D converter unit 1.	R/W
b6, b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b7	TSEN	Temperature Sensor Enable	0: Stops the temperature sensor. 1: Starts the temperature sensor.	R/W

The settings of TSCR register have the timing restrictions shown in Figure 58.4.

58.2.2 Temperature Sensor Calibration Data Register (TSCDR)

Address(es): TEMPSCONST.TSCDR FE7F 7D7Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—												
Value after reset:	0	0	0	0												

Unique value for each chip

Note: This register is only readable when the SYSCR0.ROME bit is 1 (the on-chip ROM is enabled).

The TSCDR register stores temperature sensor calibration data measured for each chip at factory shipment. The TSCDR register is a 32-bit read-only register and should be read in 32-bit units.

Temperature sensor calibration data is a digital value obtained using the 12-bit A/D converter unit 1 to convert the voltage output by the temperature sensor under the condition of $T_j = 125^\circ\text{C}$ and $AVCC1 = 3.3\text{ V}$.

The voltage $V1$ output by the temperature sensor under the condition of $T_j = 125^\circ\text{C}$ can be calculated from the value of the TSCDR register according to the formula below.

$$V1 = 3.3 \times \text{value of the TSCDR register} / 4096 \text{ [V]}$$

Note that the $AVCC1$ voltage does not affect voltage $V1$.

58.3 Using the Temperature Sensor

The temperature sensor outputs a voltage proportional to temperature. By converting this voltage into a digital value with a 12-bit A/D converter unit 1 and converting it into temperature, the internal temperature of the MCU can be obtained.

58.3.1 Preparation for Using the Temperature Sensor

Perform a calibration of the temperature sensor as shown below. The voltage output by the temperature sensor is proportional to temperature, which can be calculated according to the following formula.

Formula for the temperature characteristic:

$$T = (V_s - V_1)/\text{Slope} + T_1$$

T: Measured temperature (°C)

V_s: Voltage output by the temperature sensor when the temperature is measured (V)

T₁: Sample temperature measurement at first point (°C)

V₁: Voltage output by the temperature sensor when T₁ is measured (V)

T₂: Sample temperature measurement at second point (°C)

V₂: Voltage output by the temperature sensor when T₂ is measured (V)

Slope: Temperature slope of the temperature sensor (V/°C); Slope = (V₂ - V₁)/(T₂ - T₁)

Since there are individual differences in temperature sensors, it is recommended to prepare a temperature slope by performing trial measurements at two different temperatures as shown below.

Use the 12-bit A/D converter unit 1 to measure the voltage V₁ output by the temperature sensor at temperature T₁.

Again, using the 12-bit A/D converter unit 1, measure the voltage V₂ output by the temperature sensor at a different temperature T₂. Obtain the temperature slope (Slope = (V₂ - V₁)/(T₂ - T₁)) from these results.

Subsequently, obtain temperatures by substituting the slope into the formula for the temperature characteristic (T = (V_s - V₁)/Slope + T₁).

If you are using the temperature slope given in Table 63.57 of section 63, Electrical Characteristics, use the 12-bit A/D converter unit 1 to measure the voltage V₁ output by the temperature sensor at temperature T₁, and then calculate the temperature characteristic by using the formula below.

However, this calibration gives less accurate temperatures than two-point calibration.

$$T = (V_s - V_1)/\text{Slope} + T_1$$

T: Measured temperature (°C)

V_s: Voltage output by the temperature sensor when the temperature is measured (V)

T₁: Sample temperature measurement at first point (°C)

V₁: Voltage output by the temperature sensor when T₁ is measured (V)

Slope: Temperature slope given in Table 63.57 ÷ 1000 (V/°C)

In this MCU, the TSCDR register stores the temperature value (CAL₁₂₅) of the temperature sensor measured under the condition of T_j = 125°C and AVCC1 = 3.3 V. By using this value as the sample measurement result at the first point, preparation before using the temperature sensor can be omitted.

If V₁ is calculated from CAL₁₂₅,

$$V_1 = 3.3 \times \text{CAL}_{125}/4096 \text{ [V]}$$

Using this, the measured temperature can be calculated according to the formula below.

$$T = (V_s - V_1) / \text{Slope} + 125 \text{ [}^\circ\text{C]}$$

T: Measured temperature (internal temperature of the MCU) ($^\circ\text{C}$)

V_s : Voltage output by the temperature sensor when the temperature is measured (V)

V_1 : Voltage output by the temperature sensor when $T_j = 125^\circ\text{C}$ and $AVCC1 = 3.3 \text{ V}$ (V)

Slope: Temperature slope given in Table 63.57 $\div 1000 \text{ (V/}^\circ\text{C)}$

Error in the measured temperature (the range of variation is 3σ) is shown in Figure 58.2.

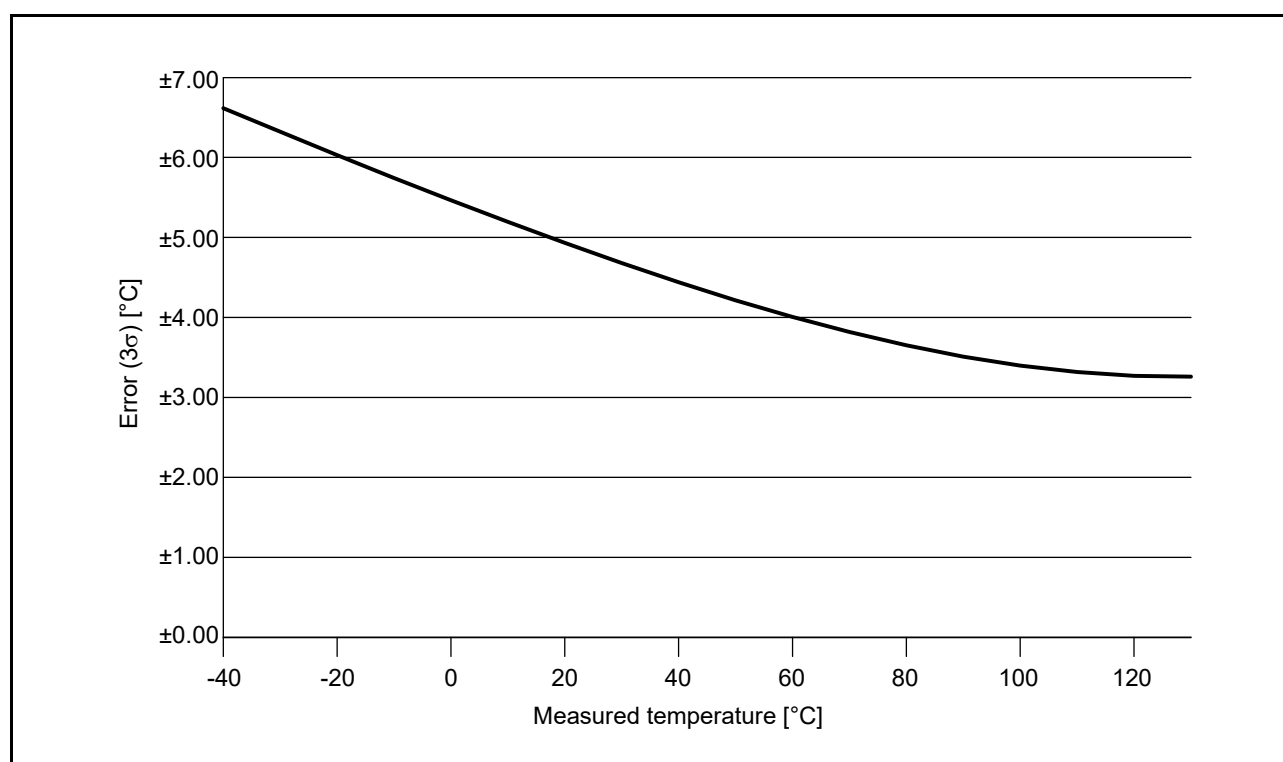


Figure 58.2 Error in the Measured Temperature

58.3.2 Setting of 12-Bit A/D Converter Unit 1

For A/D conversion of temperature sensor output voltages, 12-bit A/D converter unit 1 registers should be set as follows.

- **Selecting the Temperature Sensor Voltage as a Source for A/D Conversion**

Select the temperature sensor voltage as a source for A/D conversion by setting the temperature sensor output A/D conversion select bit in the A/D conversion extended input control register (TSSA or TSSB in the ADEXICR register) to 1.

- **Setting Scan Mode**

Select scan mode by setting the scan mode select bits in the A/D control register (ADCSR.ADCS[1:0]).

- **Setting Addition/Average Mode**

For A/D conversion of the temperature sensor output, additional or average mode is selectable. To use either additional or average mode, set the temperature sensor output A/D converted value addition mode select bit in the A/D conversion extended input control register (ADEXICR.TSSAD) to 1, and the addition count select bits in the A/D converted value addition count select register (ADADC.ADC[2:0]) to the desired number of addition.

Furthermore, clear the AVEE bit in ADADC to 0 to select addition mode; set the AVEE bit in ADADC to 1 to select average mode. In average mode, however, the ADC[2:0] bits in ADADC should not be set to 010b or 101b.

- **Setting the Sampling Time of the 12-bit A/D converter Unit 1**

The sampling time of the 12-bit A/D converter can be changed at converting the output from the temperature sensor. The initial sampling time is 11 cycles of ADCLK. To change the setting of sampling time from 11 cycles of ADCLK, set the A/D sampling state register T (ADSSTRT) while the ADCSR.ADST bit is 0.

Setting the A/D conversion start bit in the A/D control register (ADCSR.ADST) to 1 starts A/D conversion, and the result is stored in the A/D temperature sensor data register (ADTSDR). If you will be using A/D conversion of the output from the temperature sensor, do so in accord with section 58.3.3, Procedure for Using the Temperature Sensor.

58.3.3 Procedure for Using the Temperature Sensor

Figure 58.3 shows the procedure for using the temperature sensor.

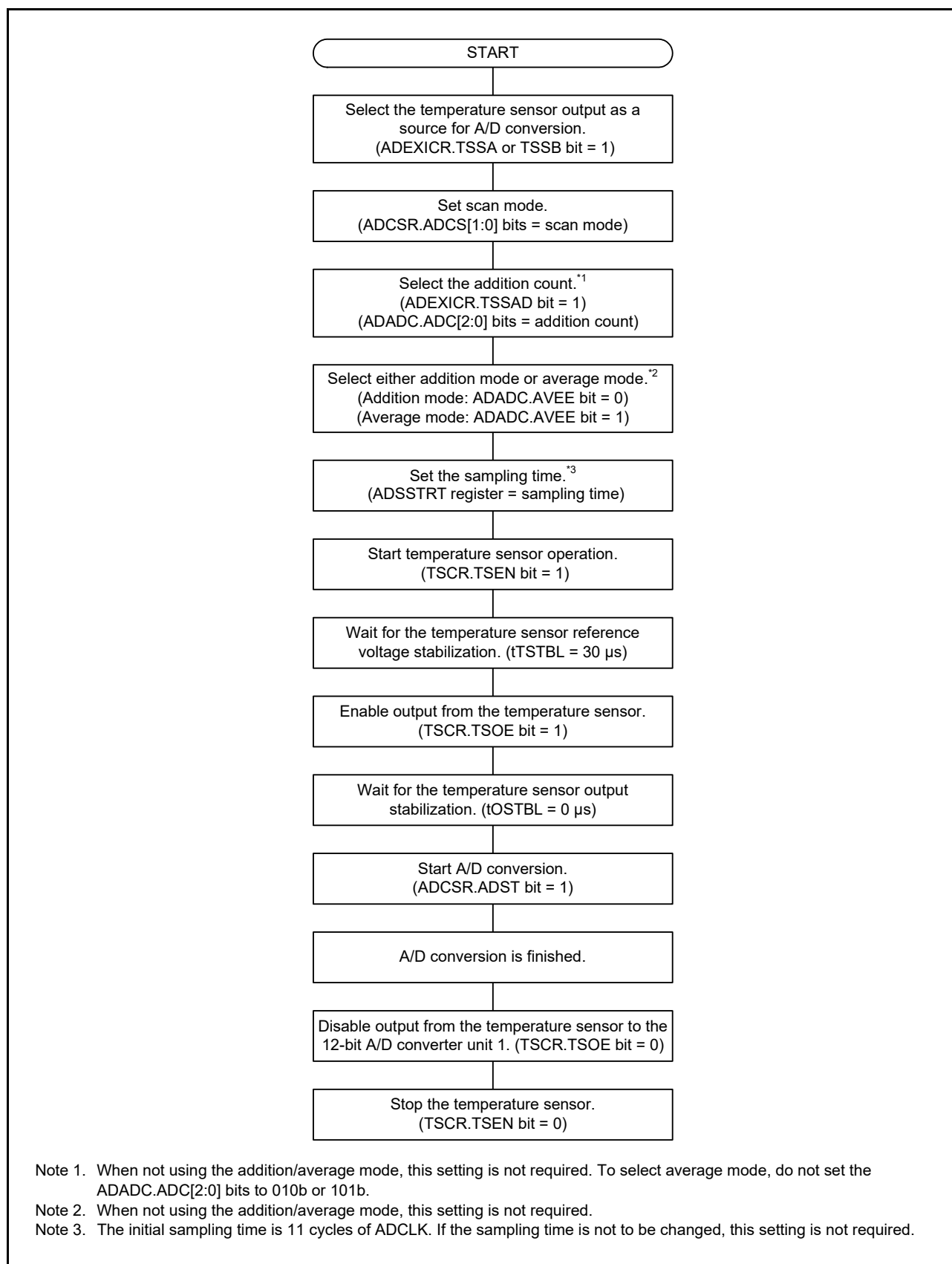


Figure 58.3 Procedure for Using the Temperature Sensor

58.3.4 Timing of A/D Conversion of Temperature Sensor Output

Figure 58.4 shows the timing from the start of temperature-sensor operation until the completion of A/D conversion when only the output from the temperature sensor is to be A/D converted and conversion is in single-scan mode. The times shown in the figure are described in Table 58.2.

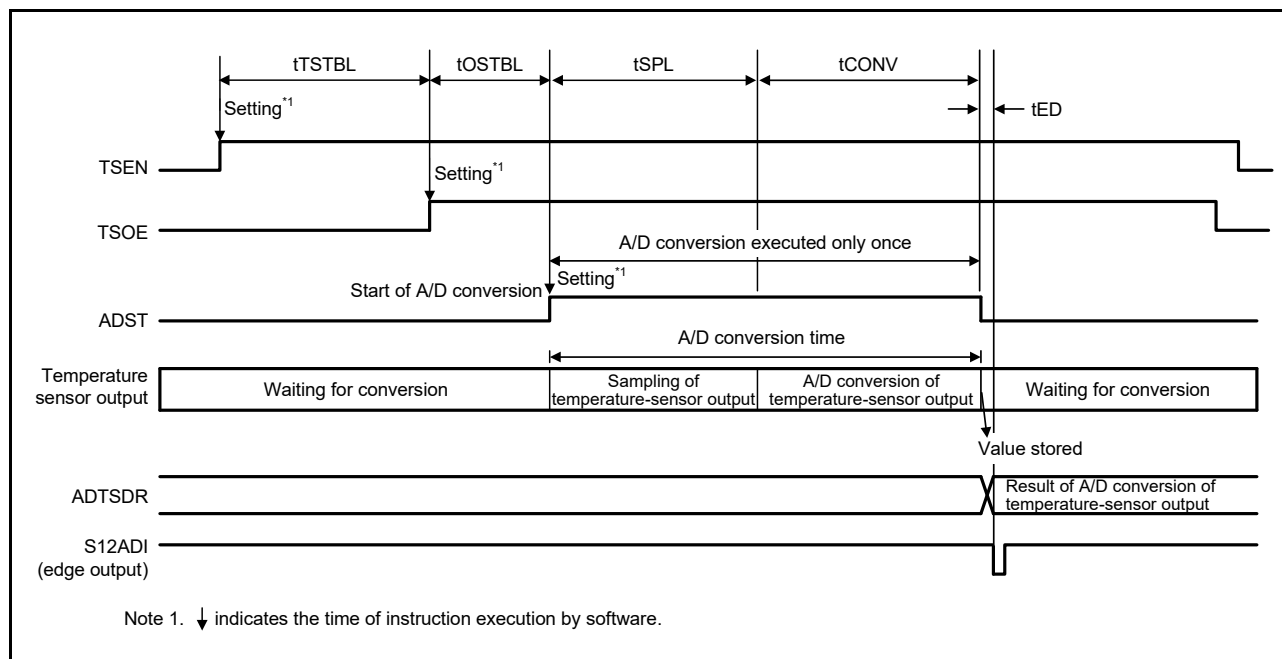


Figure 58.4 Timing from the Start of Temperature-Sensor Operation until Completion of A/D Conversion

Table 58.2 Time until Completion of A/D Conversion after the Start of Temperature-Sensor Operation

Item	Symbol	Time
Temperature-sensor reference-voltage stabilization wait time	tTSTBL	30 μ s (min)
Temperature-sensor output stabilization wait time	tOSTBL	0 μ s (min)
12-bit A/D converter unit 1 input sampling time	tSPL	ADSSTRT setting $\times$ tC(ADCLK)
A/D conversion time	tCONV	Refer to Table 56.18, Times for Conversion during Scanning (in Numbers of Cycles of ADCLK and PCLK) in section 56.3.7, Analog Input Sampling Time and Scan Conversion Time.
Scan conversion end delay time	tED	Refer to Table 56.18, Times for Conversion during Scanning (in Numbers of Cycles of ADCLK and PCLK) in section 56.3.7, Analog Input Sampling Time and Scan Conversion Time.

58.4 Usage Note

58.4.1 Module-Stop Function Setting

The corresponding bit in module stop control register B (MSTPCRB) can be used to enable and disable the temperature sensor. The initial setting is for the temperature sensor to be stopped. The register becomes accessible by releasing from the module-stop state. For details, refer to section 11, Low Power Consumption.